

Definizioni di AI literacy /1

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L'alfabetizzazione all'IA in quattro aspetti:

01. Conoscere / comprendere

02. Applicare

03. Valutare / creare l'IA

04. Comprenderne l'etica

Obiettivo:

Sviluppare le competenze necessarie per valutare criticamente l'IA. Educare gli studenti a comunicare e collaborare in modo efficace con le tecnologie intelligenti.

AI Literacy: Definition, Teaching, Evaluation and Ethical Issues

Davy Tsz Kit Ng

Faculty of Education
University of Hong Kong, China
davyngtk@connect.hku.hk

Jac Ka Lok Leung

Faculty of Education
University of Hong Kong, China
egjac@connect.hku.hk

Kai Wah Samuel Chu

Faculty of Education
University of Hong Kong, China
samchu@hku.hk

Maggie Shen Qiao

Faculty of Education
University of Hong Kong, China
sqiao@connect.hku.hk

ABSTRACT

Artificial Intelligence (AI) is at the top of the agenda for education leaders today in educating the next generation across the globe. However, public understanding of AI technologies and how to define AI literacy is under-explored. This vision poses upcoming challenges for our next generation to learn about AI. On this note, an exploratory review was conducted to conceptualize the newly emerging concept "AI literacy", in search for a sound theoretical foundation to define, teach and evaluate AI literacy. Grounded in literature on 18 existing peer-reviewed articles, this review proposed four aspects (i.e. know and understand, use, evaluate, and ethical issues) for fostering AI literacy based on the adaptation of classic literacies. This study sheds light on the consolidated definition, teaching, and ethical concerns on AI literacy, establishing the groundwork for future research such as competency development and assessment criteria on AI literacy.

KEYWORDS

AI literacy; AI learning and teaching; AI in education; AI ethics; AI literacy questionnaire.

INTRODUCTION

Artificial intelligence (AI) was first defined as "the science and engineering of making intelligent machines" in 1956 (McCarthy, 2007, p.2). Throughout several decades of the 20th century, AI has evolved progressively into intelligent machines and algorithms that can reason and adapt based on sets of rules and environment which mimic human intelligence (McCarthy, 2007). Wang (2019) broadened the definition of AI which can perform cognitive tasks particularly learning and problem-solving with the exciting technological innovations such as machine learning and neural networks (Zawacki-Richter et al., 2019). Currently, the use of AI has spread across industries (e.g., business, science, art, education) to enhance user experience and improve efficiency. Applications of AI exist in many parts of our everyday life (e.g., smart home appliances, smartphones, Google, Siri). Vast majority of the public acknowledges the existence of AI services and devices, but seldom do they know about the concepts and technology behind, or aware of potential ethical issues related to AI (Burgstainer et al., 2016; Ghallab, 2019). Similar to classic literacy which includes reading/writing and mathematical abilities, AI literacy has emerged as a new skill set in response to this new era of intelligence. As a small number of existing articles found in the Web of Science and Scopus databases, Google Scholar search is used to identify the dramatic increase in AI literacy publications from 2014 to 2021 (see Fig. 1).

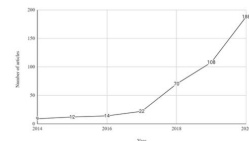


Figure 1. AI Literacy Articles from Google Scholar Published by Year

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